# SAFAR

# Phase 1

# Team 1

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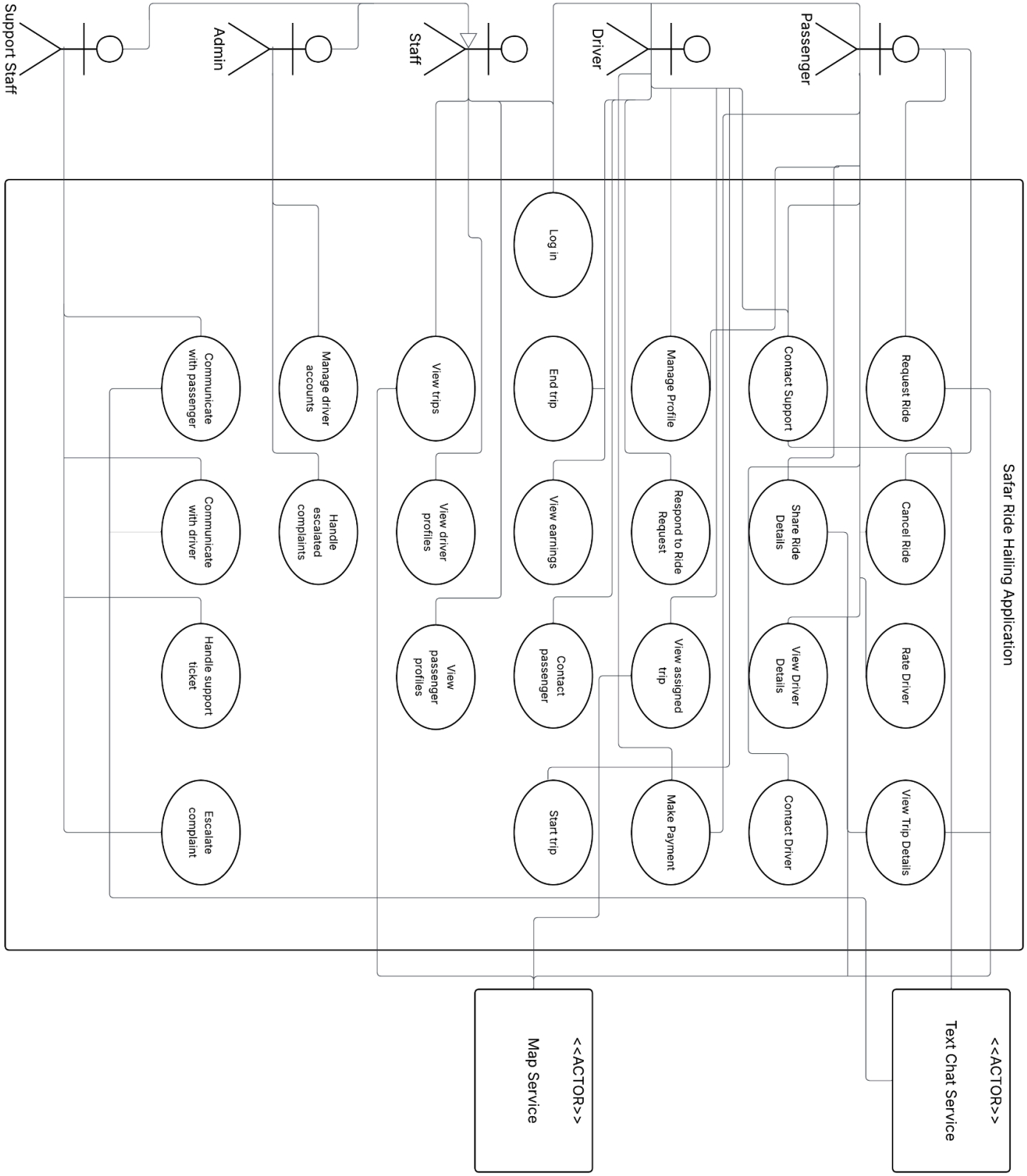
# TABLE OF CONTENTS


| | |
|---|---|
| **Use Case Diagram** | **2** |
| **Use Case Description 1** | **3** |
| **Use Case Description 2** | **4** |
| **Use Case Description 3** | **5** |
| **Use Case Description 4** | **7** |
| **Use Case Description 5** | **8** |
| **Use Case Description 6** | **9** |
| **Use Case Description 7** | **10** |
| **Use Case Description 8** | **11** |
| **Use Case Description 9** | **12** |
| **Use Case Description 10** | **13** |
| **Use Case Description 11** | **14** |
| **Use Case Description 12** | **15** |
| **Use Case Description 13** | **16** |
| **Use Case Description 14** | **18** |
| **Use Case Description 15** | **19** |
| **Use Case Description 16** | **20** |
| **Use Case Description 17** | **21** |
| **Use Case Description 18** | **22** |
| **Use Case Description 19** | **23** |
| **Use Case Description 20** | **24** |
| **Use Case Description 21** | **25** |
| **Use Case Description 22** | **27** |
| **Use Case Description 23** | **28** |
| **Use Case Description 24** | **29** |
| **Use Case Description 25** | **30** |
| **Use Case Description 26** | **31** |
| **Analysis Class Diagram** | **33** |

# Use Case Diagram



# Use Case Description 1

| | |
|---|---|
| **Identifier** | UC001 |
| **Name** | Log in |
| **Summary** | This use case describes the login procedure for users, where the user enters their credentials and is logged in by the system after verification |
| **Priority** | High |
| **Actors** | Passenger, Driver, Staff |
| **Pre-condition(s)** | None |
| **Post-condition(s)** | The user is logged in to their account |
| **Dependencies** | None |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | User selects ‘Log in’ | 1.1 | System displays login screen |
| 2 | User enters username and password and clicks confirm | 2.1 | The system authenticates the entered password |
| | | 2.2 | If username is valid and password is correct, the user is logged in successfully |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| A1 | Passenger selects ‘Sign up’ | A1.1 | System displays sign up screen |
| A2 | Passenger enters a username and | A2 | System creates a new user profile and logs the |

| | | | |
|---|---|---|---|
| | a password | .1 | user in |
| B1 | Actor tries to log in with a username that does not exist | B1 .1 | System displays an error |
| C1 | Actor tries to log in with incorrect password | C1 .1 | System displays an error |
| **Exceptions** | None | | |

# Use Case Description 2

| | |
|---|---|
| **Identifier** | UC002 |
| **Name** | Request Ride |
| **Summary** | This use case describes the procedure for a passenger to request a ride, which includes entering pickup and dropoff locations, selecting a preferred vehicle type, and confirming the ride request |
| **Priority** | High |
| **Actors** | Passenger, Driver |
| **Pre-condition(s)** | 1 - Passenger account exists and the Passenger is logged in<br>2 - Passenger's device has an active internet connection<br>3 - Location services (GPS) are enabled on the Passenger's device |
| **Post-condition(s)** | The ride request is successfully submitted and broadcasted to nearby available Drivers |
| **Dependencies** | UC001 (Log in) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger opens the application and enters the desired drop-off location | 1.1 | System detects Passenger's current location via GPS and sets it as the default pickup location |
| | | 1.2 | System calculates the route and displays available vehicle types along with estimated |

| | | | |
|---|---|---|---|
| | | | fares and arrival times for each |
| **2** | Passenger selects a preferred vehicle type and taps the confirm button | **2.1** | System validates the request |
| | | **2.2** | System displays a looking for drivers screen |
| | | **2.3** | System broadcasts the ride request to nearby available Drivers |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Passenger enters drop-off location but decides to manually change the pickup location on the map | **1.1** | System updates the map to reflect the new pickup location |
| | | **1.2** | System recalculates the route, estimated fares, and arrival times for all vehicle types based on the new locations |
| **2** | Passenger proceeds to select vehicle type and confirms | **2.1** | System continues with step 2.1 of the primary flow |
| **Exceptions** | **No Drivers Available:** System stops broadcasting, displays a "No drivers available" message, and prompts the Passenger to try again later if no Driver accepts the request within a predefined time limit<br>**Location Not Found:** System displays an error message prompting the Passenger to re-enter or manually adjust the pin on the map if the entered pickup or dropoff location cannot be pinpointed | | |

# Use Case Description 3

| | |
|---|---|
| **Identifier** | UC003 |
| **Name** | Cancel ride |
| **Summary** | This use case describes the process for a passenger to cancel a ride request, either while searching for a driver or after a driver has already accepted the trip |

| | |
|---|---|
| **Priority** | Medium |
| **Actors** | Passenger, Driver |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger has an active ride request or an assigned driver<br>3 - The trip has not yet started |
| **Post-condition(s)** | The ride is canceled, and the Driver is notified if assigned |
| **Dependencies** | UC001 (Log in), UC002 (Request ride) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger taps the cancel button on the looking for drivers screen | 1 | System displays a confirmation prompt |
| 2 | Passenger taps yes to confirm | 2.1 | System stops broadcasting the ride request to drivers |
| | | 2.2 | System updates the ride status to canceled and returns the Passenger to the home screen |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger taps the cancel ride button on the active trip screen | 1.1 | System displays a confirmation prompt |
| 2 | Passenger taps yes to confirm | 2.1 | System cancels the active trip and immediately notifies the assigned Driver |
| | | 2.3 | System returns the Passenger to the home screen |

| | |
|---|---|
| **Exceptions** | **Ride Already Started:** System disables the cancel ride button and displays a message advising the Passenger to ask the Driver to end the trip early or contact Support staff if the Driver has already swiped to start the trip |

# Use Case Description 4

| | |
|---|---|
| **Identifier** | UC004 |
| **Name** | Rate driver |
| **Summary** | This use case describes the process for a passenger to submit a rating for a driver after a ride is completed |
| **Priority** | Low |
| **Actors** | Passenger |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Driver has successfully ended the trip |
| **Post-condition(s)** | The rating and feedback are saved in the system and linked to the Driver's profile |
| **Dependencies** | UC001 (Log in) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger views the rating prompt displayed after the trip ends | 1.1 | System displays the driver's name, photo, and a 1 to 5 star rating scale |
| 2 | Passenger selects a star rating | 2.1 | System highlights the selected stars and enables the submit button |
| 3 | Passenger types optional feedback and taps the submit button | 3.1 | System saves the rating to the database |
| | | 3.2 | System displays a brief thank you confirmation |
| | | 3.3 | System redirects the Passenger to the home screen |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | Passenger decides not to leave a rating and taps the skip button | 1.1 | System dismisses the rating prompt |
| | | 1.2 | System redirects the Passenger to the home screen |
| **Exceptions** | | | |

# Use Case Description 5

| | |
|---|---|
| **Identifier** | UC005 |
| **Name** | Contact support |
| **Summary** | This use case describes the process for a passenger to reach out to support staff for assistance regarding a ride, account issues, or general inquiries |
| **Priority** | Medium |
| **Actors** | Passenger, Support staff |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger's device has an active internet connection |
| **Post-condition(s)** | A support ticket or chat session is successfully created and routed to Support staff |
| **Dependencies** | UC001 (Log in) |

| **Typical Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger navigates to the help section in the application | 1 | System displays a list of common issues and a contact support button |
| 2 | Passenger taps the contact support button and selects the option to call support staff | 2 | System initiates a call with a support staff member |
| **Alternate Course of Action** | | | |

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger chooses the live chat option instead of audio call | 1 | System opens a chat window and attempts to connect with Support staff |
| 2 | Passenger types a message and sends it | 2 | System connects the Passenger with Support staff and displays the real-time conversation |
| **Exceptions** | **Support Offline:** System informs the Passenger that Support staff are currently offline and prompts them to submit a ticket instead of initiating a live chat if outside operating hours | | |

# Use Case Description 6

| | |
|---|---|
| **Identifier** | UC006 |
| **Name** | View trip details |
| **Summary** | This use case describes the process for a passenger to view information about an assigned trip including the driver's information, estimated time of arrival (ETA), and estimated fare |
| **Priority** | High |
| **Actors** | Passenger |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger has an active ride that has been accepted by a Driver |
| **Post-condition(s)** | Passenger successfully views the real-time details and status of their active trip |
| **Dependencies** | UC001 (Log in), UC002 (Request ride) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger views the active trip screen on the application | 1.1 | System displays the assigned Driver's name, photo, vehicle model, and license plate |
| | | 1.2 | System calculates and displays the estimated |

| | | | |
|---|---|---|---|
| | | | time of arrival (ETA) to the pickup or dropoff location |
| | | **1.3** | System calculates and displays the estimated fare for the trip |
| **2** | Passenger taps the screen to refresh the map view | **2.1** | System updates the Driver's real-time location on the map and adjusts the ETA accordingly |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Passenger taps the estimated fare amount to see more information | **1** | System displays a detailed breakdown of the estimated fare including base rate and distance charges |
| **2** | Passenger dismisses the fare breakdown | **2** | System returns the Passenger to the main active trip screen |
| **Exceptions** | **Location Tracking Failure:** System displays a message indicating the Driver's real-time location is temporarily unavailable and shows the last known ETA if the Driver's GPS signal is lost | | |

# Use Case Description 7

| | |
|---|---|
| **Identifier** | UC007 |
| **Name** | Share ride details |
| **Summary** | This use case describes the process for a passenger to share their live trip status, driver information, and estimated time of arrival with a contact |
| **Priority** | Low |
| **Actors** | Passenger |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger has an active ongoing trip |
| **Post-condition(s)** | A tracking link containing the live trip details is successfully generated and sent to the selected contact |

| | | | |
|---|---|---|---|
| **Dependencies** | UC001 (Log in), UC002 (Request ride) | | |
| **Typical Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger taps the share ride button on the active trip screen | 1 | System generates a unique tracking link for the current trip |
| 2 | Passenger copies the message and returns to trip screen | 2 | System resumes displaying the active trip screen |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | None | 1 | None |
| **Exceptions** | **No Active Trip:** System disables or hides the share ride button if the trip has already ended or been canceled | | |

# Use Case Description 8

| | |
|---|---|
| **Identifier** | UC008 |
| **Name** | View driver details |
| **Summary** | This use case describes the process for a passenger to view the profile and detailed information of their assigned driver |
| **Priority** | Low |
| **Actors** | Passenger |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger has an assigned driver for an active trip |
| **Post-condition(s)** | Passenger successfully views the driver's profile information including ratings and vehicle details |
| **Dependencies** | UC001 (Log in), UC002 (Request ride) |
| **Typical Course of Action** | |

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger taps on the driver's photo or name on the active trip screen | 1 | System displays the driver's profile including name, photo, overall rating, number of trips, and full vehicle details |
| 2 | Passenger reviews the information and taps the close or back button | 2 | System closes the driver profile and returns the Passenger to the active trip screen |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger taps on the driver's rating to view more information | 1 | System displays a list of recent anonymous feedback or badges earned by the driver |
| 2 | Passenger scrolls through the feedback and taps the back button | 2 | System returns the Passenger to the main driver profile screen |
| **Exceptions** | | | |

# Use Case Description 9

| | |
|---|---|
| **Identifier** | UC009 |
| **Name** | Contact driver |
| **Summary** | This use case describes the process for a passenger to communicate with their assigned driver via in-app chat or phone call regarding pickup details or delays |
| **Priority** | High |
| **Actors** | Passenger, Driver |
| **Pre-condition(s)** | 1 - Passenger is logged in<br>2 - Passenger has an assigned Driver for an active trip |
| **Post-condition(s)** | Passenger successfully sends a message or initiates a phone call to the assigned Driver |
| **Dependencies** | UC001 (Log in), UC002 (Request ride) |

| Typical Course of Action | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger taps the chat icon on the active trip screen | 1 | System opens the in-app messaging interface |
| 2 | Passenger types a message and taps the send button | 2.1 | System transmits the message to the assigned Driver and displays it in the chat window |
| | | 2.2 | System updates the message status to delivered or read when applicable |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger taps the phone icon on the active trip screen instead of chat | 1 | System prompts the Passenger to confirm they want to call the Driver |
| 2 | Passenger taps call to confirm | 2 | System initiates the call between the Passenger and Driver |
| **Exceptions** | **Ride Canceled or Ended:** System disables the chat and call icons and displays a message that the Driver can no longer be contacted if the trip is completed or canceled | | |

# Use Case Description 10

| | |
|---|---|
| **Identifier** | UC010 |
| **Name** | Manage profile |
| **Summary** | This use case describes the procedure for a driver to view and update their profile information, including personal details, vehicle information, and profile picture. |
| **Priority** | Medium |
| **Actors** | Driver, Passenger |

| | |
|---|---|
| **Pre-condition(s)** | 1. The actor's account exists and the actor is logged in<br>2. The actor's device has an active internet connection |
| **Post-condition(s)** | The actor's profile is successfully updated and saved |
| **Dependencies** | UC001(Log In) |

| **Typical Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Actor navigates to the profile section | 1.1 | System loads and displays the actor's current profile information |
| 2 | Actor edits the desired fields and taps 'Save' | 2.1 | System validates the updated information |
| | | 2.2 | System saves the changes and displays a success confirmation message |

| **Alternate Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Actor enters invalid information | 1.1 | Actor displays an error message |

| | |
|---|---|
| **Exceptions** | None |

# Use Case Description 11

| | |
|---|---|
| **Identifier** | UC011 |
| **Name** | Respond to Ride Request |
| **Summary** | This use case describes the procedure for a driver to view an incoming ride request and either accept or reject it. |
| **Priority** | High |
| **Actors** | Driver, Passenger |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection. |

| | |
|---|---|
| | 3. The Driver is online and available to accept rides |
| **Post-condition(s)** | The Driver has either accepted or rejected the ride request. If accepted, the Passenger is notified and trip begins |
| **Dependencies** | UC001 (Log In), UC002 (Request Ride) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver receives and views the incoming ride request | 1.1 | System displays ride request details including passenger pickup location, dropoff location and estimated fare |
| 2 | Driver taps "Accept" | 2.1 | System confirms the acceptance and notifies the Passenger |
| | | 2.2 | System displays the Passenger's pickup location on the map for the Driver |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver taps "Reject" | 1.1 | System cancels the request for this Driver and broadcasts it to other available Drivers |

| | |
|---|---|
| **Exceptions** | **Drivers unavailable:** System notifies the Passenger that no drivers are available |

# Use Case Description 12

| | |
|---|---|
| **Identifier** | UC012 |
| **Name** | View Assigned Trip |
| **Summary** | This use case describes the procedure for a driver to view the assigned trip details including the passenger's pickup and dropoff locations after accepting a ride request. |
| **Priority** | High |

| | |
|---|---|
| **Actors** | Driver |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection.<br>3. The Driver has accepted a ride request. |
| **Post-condition(s)** | The Driver can clearly view the passenger's pickup and dropoff locations on the map. |
| **Dependencies** | UC001 (Log In), UC011 (Respond to Ride Request) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver navigates to the current trip screen | 1.1 | System displays the passenger's pickup and dropoff locations on the map |
| | | 1.2 | System displays the recommended route and estimated arrival time to the pickup location |
| 2 | Driver reviews the trip details | 2.1 | System keeps the map updated in real time as the Driver moves |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver loses internet connection while viewing trip details | 1.1 | System displays an error message |

| | |
|---|---|
| **Exceptions** | None. |

# Use Case Description 13

| | |
|---|---|
| **Identifier** | UC013 |
| **Name** | Start Trip |
| **Summary** | This use case describes the procedure for a driver to start the trip after arriving at the passenger's pickup location and confirming the passenger's presence. |

| | |
|---|---|
| **Priority** | High |
| **Actors** | Driver, Passenger |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection.<br>3. The Driver has accepted a ride request and arrived at the pickup location |
| **Post-condition(s)** | The trip is successfully started and both the Driver and Passenger are notified. |
| **Dependencies** | UC001 (Log In), UC012 (View Assigned Trip) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver arrives at the passenger's pickup location and taps "Start Trip" | 1.1 | System verifies that pickup location is correct |
| | | 1.2 | System starts the trip and begins tracking the route |
| | | 1.3 | System notifies the Passenger that the trip has started |
| 2 | Driver begins driving towards the dropoff location | 2.1 | System tracks and updates the Driver's location in real time |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver taps "Start Trip" but is at wrong pickup location | 1.1 | System shows  an error message |

| | |
|---|---|
| **Exceptions** | None |

# Use Case Description 14

| | |
|---|---|
| **Identifier** | UC014 |
| **Name** | End Trip |
| **Summary** | This use case describes the procedure for a driver to end the trip after arriving at the passenger's dropoff. |
| **Priority** | High |
| **Actors** | Driver, Passenger |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection.<br>3. A trip is currently in progress |
| **Post-condition(s)** | The trip is successfully completed ,fare is calculated and both the Driver and Passenger are notified. |
| **Dependencies** | UC001 (Log In), UC013 (Start Trip), UC026 (Make Payment) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver arrives at the passenger's dropoff location and taps "End Trip" | 1.1 | System verifies that dropoff location is correct |
| | | 1.2 | System ends the trip and calculate the final fare |
| | | 1.3 | System notifies the Passenger that the trip has ended |
| 2 | Driver views the trip summary | 2.1 | System displays the completed trip details including distance,fare,duration |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver taps "Ends Trip" but is at wrong dropoff location | 1.1 | System shows an error message |

| | | | |
|---|---|---|---|
| 2 | Driver confirms ending the trip at current location | | System ends the trip and calculates fare based on actual distance travelled |
| **Exceptions** | None | | |

# Use Case Description 15

| | |
|---|---|
| **Identifier** | UC015 |
| **Name** | View Earnings |
| **Summary** | This use case describes the procedure for a driver to end the trip after arriving at the passenger's dropoff. |
| **Priority** | Medium |
| **Actors** | Driver |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection.<br>3. The Driver has at least one completed trip. |
| **Post-condition(s)** | The Driver successfully views their earnings summary. |
| **Dependencies** | UC001 (Log In), UC014 (End Trip) |

| **Typical Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Driver navigates to the Earnings section | 1.1 | System retrieves and displays the Driver's total earnings |
| | | 1.2 | System displays a breakdown of earnings per completed trip |
| 2 | Driver selects a specific trip to view details | 2.1 | System displays the detailed fare breakdown for selected trip |
| **Alternate Course of Action** | | | |

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver has no completed trip | 1.1 | System displays a message to Driver informing that no earnings are available yet |
| **Exceptions** | None | | |

# Use Case Description 16

| | |
|---|---|
| **Identifier** | UC016 |
| **Name** | Contact Passenger |
| **Summary** | This use case describes the procedure for a driver to connect the Passenger via call or text during a request or trip. |
| **Priority** | Medium |
| **Actors** | Driver, Passenger |
| **Pre-condition(s)** | 1. The Driver account exists and the Driver is logged in.<br>2. The Driver's device has an active internet connection.<br>3. The Driver has an accepted or ongoing trip with the Passenger. |
| **Post-condition(s)** | The Driver has successfully connected the Passenger via preferred communication method. |
| **Dependencies** | UC001 (Log In), UC011 (Respond to Ride Request) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver navigates to the trip screen and select ‘Contact Passenger’ | 1.1 | System displays the available Contact options including call and text |
| 2 | Driver selects preferred contact method | 2.1 | System initiates the call or text interface with Passenger |

| | | | |
|---|---|---|---|
| **3** | Driver communicates with the Passenger | **3.1** | System logs the conversation for record |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Passenger does not respond to call | **1.1** | System notifies Driver that Passenger is unavailable and suggests sending text instead |
| **Exceptions** | None | | |

# Use Case Description 17

| | |
|---|---|
| **Identifier** | UC017 |
| **Name** | View trips |
| **Summary** | This use case describes how a staff member can view completed, cancelled and ongoing trips as well as their details |
| **Priority** | Medium |
| **Actors** | Staff |
| **Pre-condition(s)** | 1. Actor is logged in<br>2. Actor's device has an active internet connection |
| **Post-condition(s)** | Actor views a list of trips, and views the details of selected trip(s) |
| **Dependencies** | UC001 (Log in) |

| | | | |
|---|---|---|---|
| **Typical Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor navigates to the 'Trips' tab | **1.1** | System displays a list of all completed, |

| | | | |
|---|---|---|---|
| | | | ongoing and cancelled rides |
| 2 | Actor selects a ride to view | 2.1 | System displays all details of that specific trip, such as pickup and dropoff locations, ride duration, driver profile and passenger profile |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Actor searches for specific trip using the search bar | 1.1 | System returns a list of matching trips and typical course of action resumes |
| 2 | Actors filters trips based on specific attributes such as date or status | 2.1 | System returns a list of all trips that match those attributes and typical course of action resumes |
| **Exceptions** | None | | |

# Use Case Description 18

| | |
|---|---|
| **Identifier** | UC018 |
| **Name** | View driver profiles |
| **Summary** | This use case describes the procedure for admins and support staff to view the profiles of registered drivers |
| **Priority** | Medium |
| **Actors** | Staff |
| **Pre-condition(s)** | 1. Actor is logged in<br>2. Actor has an active internet connection |
| **Post-condition(s)** | Actor views a list of all registered drivers, and views the details of selected driver(s) |
| **Dependencies** | UC001 (Log in) |
| **Typical Course of Action** | |
| **S #** / **Actor Action** | **S #** / **System Response** |

| | | | |
|---|---|---|---|
| **1** | Actor clicks 'Drivers' | **1.1** | System provides a list of all registered drivers |
| **2** | Actor selects a specific driver | **2.1** | System displays details of driver's profile, such as driver name and vehicle type |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor searches for specific driver using the search bar | **1.1** | System returns a list of matching driver profiles and typical course of action resumes |
| **2** | Actors filters driver based on specific attributes such as vehicle type | | System returns a list of all drivers that match those attributes and typical course of action resumes |
| **Exceptions** | None | | |

# Use Case Description 19

| | |
|---|---|
| **Identifier** | UC019 |
| **Name** | View passenger profiles |
| **Summary** | This use case describes the procedure for admins and support staff to view profile details of passengers |
| **Priority** | Medium |
| **Actors** | Staff |
| **Pre-condition(s)** | 1. Actor is logged in<br>2. Actor has an active internet connection |
| **Post-condition(s)** | Actor views a list of all passengers, and views the details of selected passenger(s) |
| **Dependencies** | UC001 (Log in) |

| **Typical Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor clicks 'Passengers' | **1.1** | System provides a list of all passenger profiles |

| | | | |
|---|---|---|---|
| **2** | Actor selects a specific profile | **2.1** | System displays details of passenger's profile |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor searches for passenger using search box | **1.1** | System returns a list of matching passenger profiles, and typical course of action resumes |
| **2** | Actor filters passengers based on specific attributes | **2.1** | System returns a list of all passengers that match those attributes/conditions and typical course of action resumes |
| **Exceptions** | None | | |

# Use Case Description 20

| | |
|---|---|
| **Identifier** | UC020 |
| **Name** | Manage driver accounts |
| **Summary** | This use case describes how an admin can manage the profiles of registered drivers, which includes creating and deleting driver accounts |
| **Priority** | High |
| **Actors** | Primary: Admin<br>Secondary: Driver |
| **Pre-condition(s)** | 1. Actor is logged in<br>2. Actor has an active internet connection |
| **Post-condition(s)** | Driver account information is created/updated/deleted successfully |
| **Dependencies** | UC001 (Log In), UC018 (View Driver Profiles) |

| | | | |
|---|---|---|---|
| **Typical Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Admin selects 'Drivers' | **1.1** | System displays a list of driver profiles |
| **A1** | Admin selects 'New Driver' | **A1 .1** | System displays a new driver form |

| | | | |
|---|---|---|---|
| **A2** | Admin enters required information and submits form | **A2.1** | System validates information |
| | | **A2.2** | If info is valid, system creates new driver profile and displays confirmation message |
| **B1** | Admin selects a profile | **B1.1** | System displays driver profile |
| **B2** | Admin selects 'Edit' | **B2.1** | System displays editable driver profile |
| **B3** | Admin makes necessary changes to profile and submits | **B3.1** | System validates new information |
| | | **B3.2** | If all new information is valid, system updates driver information and displays confirmation message |
| **C1** | Admin selects 'Delete' on a profile from drivers list | **C1.1** | System asks for deletion confirmation |
| **C2** | Admin confirms deletion | **C2.1** | System deletes selected driver profile and prints confirmation message |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **A1** | Admin selects 'Delete' on a profile but cancels deletion | **A1.1** | System aborts deletion |
| **B1** | Admin leaves some required field empty while creating/editing a profile | **B1.1** | System displays error message and a prompt to fill missing information |
| **Exceptions** | None | | |

# Use Case Description 21

| | |
|---|---|
| **Identifier** | UC021 |
| **Name** | Handle Escalated Complaints |

| | |
|---|---|
| **Summary** | This use case describes how an admin can respond to complaints escalated by support staff, which includes reviewing profiles and trips, taking administrative action and resolving complaints |
| **Priority** | Medium |
| **Actors** | Admin |
| **Pre-condition(s)** | 1. Admin is logged in<br>2. Admin has an active internet connection<br>3. Support staff has escalated complaint(s) |
| **Post-condition(s)** | The admin has viewed the complaint and marked it as resolved |
| **Dependencies** | UC001 (Log in), UC018 (View driver profiles), UC019 (View passenger profiles), UC020 (Manage driver accounts), UC025 (Escalate Complaint) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Actor selects ‘Complaints’ | 1.1 | System displays a list of complaints |
| 2 | Actor selects a complaint to view | 2.1 | System displays details of complaint such as driver profile, passenger profile and complaint description |
| 3 | Actor marks complaint as resolved | 3.1 | System updates complaint status and displays confirmation |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | After viewing complaint, actor views the related driver profile | 1.1 | System follows typical course of action for UC018 |
| 2 | After viewing complaint, actor views the related passenger profile | 2.1 | System follows typical course of action for UC019 |
| 3 | After viewing complaint, actor edits or deletes the related driver’s profile | 3.1 | System follows typical course of action for UC020 (section B or C) |

| | |
|---|---|
| **Exceptions** | **Passenger or driver profile no longer exists:** System displays an error message |

# Use Case Description 22

| | |
|---|---|
| **Identifier** | UC022 |
| **Name** | Communicate with passenger |
| **Summary** | This use case describes how support staff can contact passengers, which may be done as a response to their complaints |
| **Priority** | Medium |
| **Actors** | Primary: Support staff<br>Secondary: Passenger |
| **Pre-condition(s)** | 1. Support staff is logged in<br>2. Support staff has an active internet connection<br>3. Passenger has submitted a complaint |
| **Post-condition(s)** | Support staff has successfully contacted passenger via phone call |
| **Dependencies** | UC001 (Log in), UC024 (Handle support ticket) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Support staff navigates to a complaint (UC024) | 1.1 | System displays complaint details (after following typical course of action for UC024) |
| 2 | Support staff selects ‘Contact passenger’ | 2.1 | System dials passenger’s phone number |
| 3 | Support staff calls passenger | 3.1 | None |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Passenger does not respond and support staff tries calling again | 1.1 | System continues with typical course of action |

| | |
|---|---|
| **Exceptions** | **Passenger does not respond:** After multiple unsuccessful attempts, the complaint remains unresolved |

# Use Case Description 23

| | |
|---|---|
| **Identifier** | UC023 |
| **Name** | Communicate with driver |
| **Summary** | This use case describes how support staff can contact drivers, which may be done as a response to their complaints |
| **Priority** | Medium |
| **Actors** | Primary: Support staff<br>Secondary: Driver |
| **Pre-condition(s)** | 1. Support staff is logged in<br>2. Support staff has an active internet connection<br>3. Driver has submitted a complaint |
| **Post-condition(s)** | Support staff has successfully contacted driver via phone call |
| **Dependencies** | UC001 (Log in), UC024 (Handle support ticket) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Support staff navigates to a complaint (UC024) | 1.1 | System displays complaint details (after following typical course of action for UC024) |
| 2 | Support staff selects ‘Contact Driver’ | 2.1 | System dials driver’s phone number |
| 3 | Support staff calls driver | 3.1 | None |

**Alternate Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| 1 | Driver does not respond and support staff tries calling again | 1.1 | System continues with typical course of action |

| | |
|---|---|
| **Exceptions** | **Driver does not respond:** After multiple unsuccessful attempts, the complaint remains unresolved |

# Use Case Description 24

| | |
|---|---|
| **Identifier** | UC024 |
| **Name** | Handle support ticket |
| **Summary** | This use case describes how support staff can create, edit, delete and resolve support tickets, which are made after a passenger or driver contacts support staff. |
| **Priority** | High |
| **Actors** | Support Staff |
| **Pre-condition(s)** | 1. Actor is logged in<br>2. Actor has an active internet connection<br>3. A passenger or driver has contacted support staff to register a complaint |
| **Post-condition(s)** | A support ticket has been created, edited, deleted or marked as resolved successfully |
| **Dependencies** | UC001 (Log in), UC005 (Contact support) |

**Typical Course of Action**

| S # | Actor Action | S # | System Response |
|---|---|---|---|
| **1** | Actor selects ‘Complaints’ | **1.1** | System displays a list of complaints |
| **A1** | Actor selects ‘New Ticket’ | **A1.1** | System displays a ticket form containing fields such as passenger, driver, trip details |
| **A2** | Actor fills form and submits | **A2.1** | System creates new complaint ticket and displays confirmation message |
| **B1** | Actor selects ‘Delete’ on target ticket | **B1.1** | System displays prompt to confirm deletion |
| **B2** | Actor confirms deletion | **B2.1** | System deletes ticket and displays confirmation message |

| | | | |
|---|---|---|---|
| **C1** | Actor selects 'Mark as Resolved' on target ticket | **C1.1** | System displays prompt requesting confirmation |
| **C2** | Actor confirms action | **C2.1** | System marks ticket as resolved and displays confirmation |
| **D1** | Actor selects 'Edit' on target ticket | **D1.1** | System displays the filled ticket form, prompting updation |
| **D2** | Actor makes necessary changes and submits | **D2.1** | System saves updated version of complaint ticket and displays confirmation message |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor leaves field(s) in ticket form missing (while creating or editing ticket) | **1.1** | System displays an error message and returns to ticket, prompting actor to fill the required fields |
| **2** | Actor searches for a ticket instead of selecting it from the list | **2.1** | System returns all tickets that match the search, and typical course of action resumes |
| **3** | Actor filters out tickets based on certain conditions before selecting one | **3.1** | System returns a list of all tickets that match the entered criteria, and typical course of action resumes |
| **Exceptions** | **Action cancelled:** System returns and the requested action is not performed | | |

# Use Case Description 25

| | |
|---|---|
| **Identifier** | UC025 |
| **Name** | Escalate complaint |
| **Summary** | This use case describes how support staff can escalate support tickets to admins |
| **Priority** | Medium |
| **Actors** | Support staff |
| **Pre-condition(s)** | 1. Actor is logged in |

| | |
|---|---|
| | 2. Actor has an active internet connection<br>3. A support ticket exists |
| **Post-condition(s)** | The target support ticket has been forwarded to admins for review |
| **Dependencies** | UC001 (Log in), UC024 (Handle support ticket) |

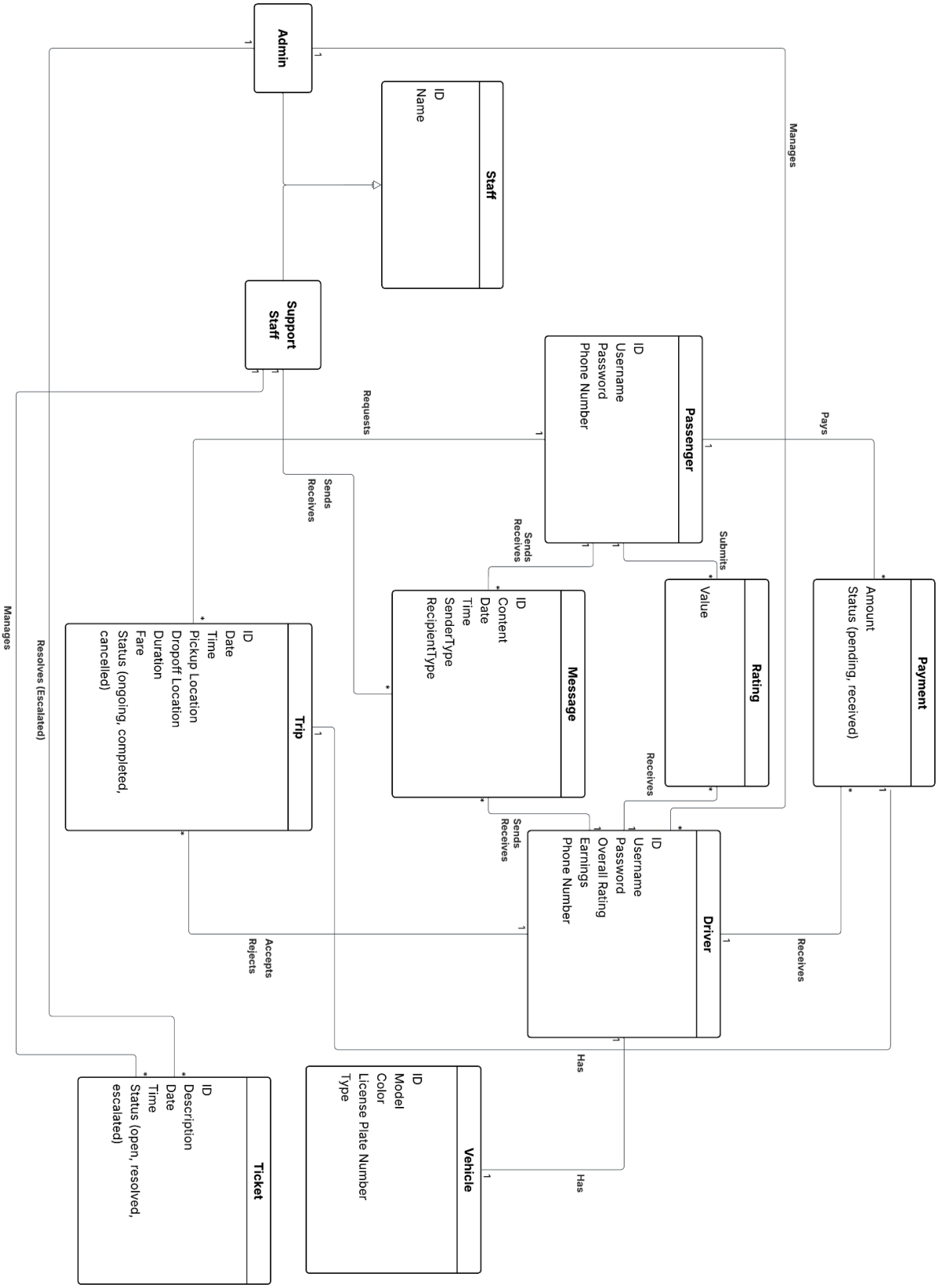
| **Typical Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| **1** | Actor selects 'Complaints' | **1.1** | System displays a list of tickets |
| **2** | Actor selects 'Escalate' on target complaint | **2.1** | System displays a prompt asking for confirmation |
| **3** | Actor confirms escalation | **3.1** | System forwards complaint to admins and displays confirmation message |

| **Alternate Course of Action** | | | |
|---|---|---|---|
| **S #** | **Actor Action** | **S #** | **System Response** |
| | None | | None |

| | |
|---|---|
| **Exceptions** | **Escalation not confirmed:** System does not escalate the ticket to admins |

# Use Case Description 26

| | |
|---|---|
| **Identifier** | UC026 |
| **Name** | Make payment |
| **Summary** | This use case describes how a passenger pays the driver and the driver marks fare as paid |
| **Priority** | High |
| **Actors** | Driver, Passenger |
| **Pre-condition(s)** | 1. Driver is logged in<br>2. Driver has an active internet connection |

| | | | |
|---|---|---|---|
| **Post-condition(s)** | The fare of the trip will be paid | | |
| **Dependencies** | UC001 (Log in) | | |
| **Typical Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| 1 | Passenger pays fare to driver and driver selects 'Payment Received' | 1.1 | System marks the payment as complete |
| **Alternate Course of Action** | | | |
| **S #** | **Actor Action** | **S #** | **System Response** |
| | None | | None |
| **Exceptions** | None | | |

# Analysis Class Diagram